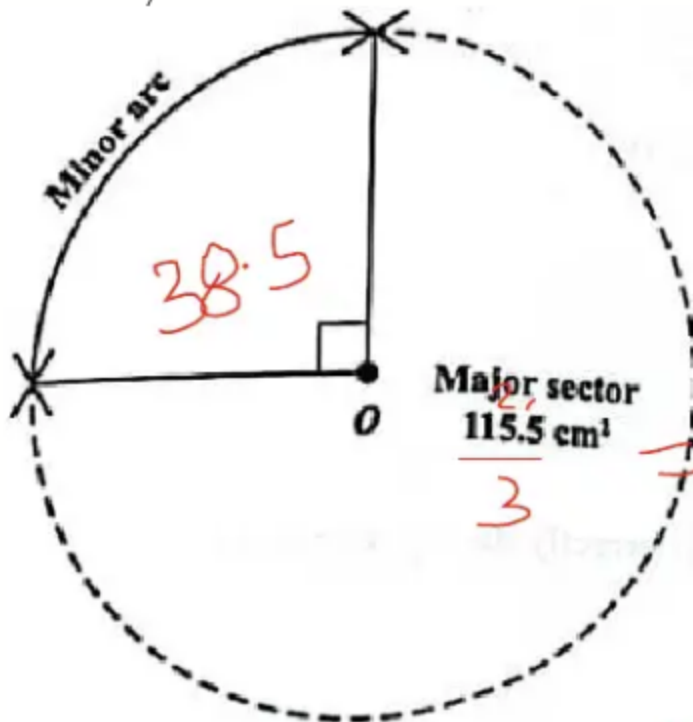


CSEC January 2025 Mathematics Paper 1

- 1) What percentage of 40 is 8?
A. 5%
B. 20%
C. 32%
D. 150%
- 2) The value of the digit 2 in the number 748.621 is
A. $\frac{2}{10^2}$
B. $\frac{2}{10^3}$
C. 2×10^{-1}
D. 2000×10^{-3}
- 3) $\sqrt{17^2 - 15^2}$ is equal to
A. 1
B. 2
C. 8
D. 16
- * 4) In a school, the ratio of the number of pupils to the number of teachers is 20:1. If the number of pupils is 840, how many teachers are there?
A. 40
B. 42
C. 820
D. 840
Handwritten notes: PT, 1x42, 840 ÷ 20 = 42
- 5) A bag of apples can be shared equally among either 6, 10 or 15 children. The MINIMUM number of apples that is likely to be in the bag is
A. 30
B. 31
C. 60
D. 90
- 6) 99×101 has the same value as
A. $(99 \times 100) + 1$
B. $(99 \times 100) (99 \times 1)$
C. $(99 \times 100) - (99 \times 1)$
D. $(99 \times 100) + (99 \times 1)$

- * 36) The following diagram shows a circle, centered at O. the major sector and minor are indicated.
(Use $\pi = \frac{22}{7}$)



$$\frac{1}{4} \pi r^2 = 38.5$$

$$115.5 + 38.5 = 154$$

$$\pi r^2 \times 154 = 0$$

If the area of the major sector is 115.5 cm^2 , then the DIAMETER of the circle, in cm, is

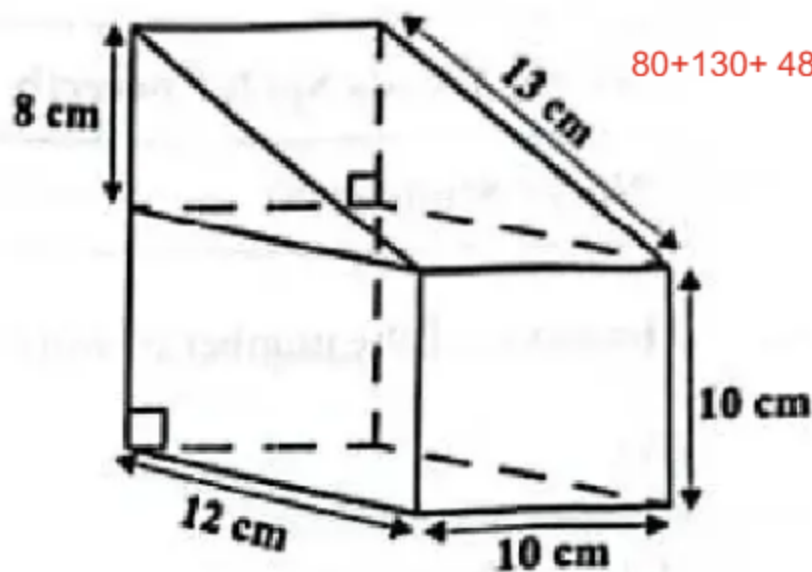
- A. 5.5
- B. 7.0
- ☒ C. 14.0
- D. 28.0

$$\frac{22}{7} \times 7 \times 7 = 154$$

$$r = 7$$

$$D =$$

- 37) The following diagram shows a compound solid.



$$80 + 130 + 48 + 48 + 100 + 100 + 120 + 120 + 120$$

$$80 + 130 + 96 + 200 + 360 = 770 + 96 = 866$$

The surface area of the compound solid, in cm^2 , is

- A. 746
 - B. 866
 - C. 980
 - D. 1100
- 38) The distance around a lake is 8km. On a map, this distance around the lake is represented by a length of 4cm. The scale on the map is
- A. 1 : 40
 - B. 1 : 2000
 - C. 1 : 100000
 - D. 1 : 200000
- 39) The following table shows the number of words that a group of 60 students got correct in a spelling test that consisted of 10 words.

No. of Words Spelt Correctly	6	7	8	9	10
No. of Students	3	18	7	16	16

The mode of the number of words spelt correctly is

- A. 7
- B. 8
- C. 16
- D. 18

- 40) The following table shows the number of words that a group of 60 students got correct in a spelling test that consisted of 10 words.

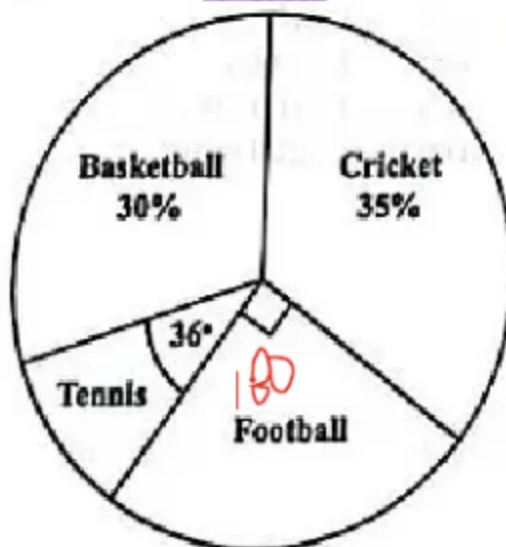
No. of Words Spelt Correctly	6	7	8	9	10
No. of Students	3	18	7	16	16

$$61/2 = 30.5$$

$$30\text{th} + 31\text{st} / 2$$

The median number of words the student spelt correctly during the test is

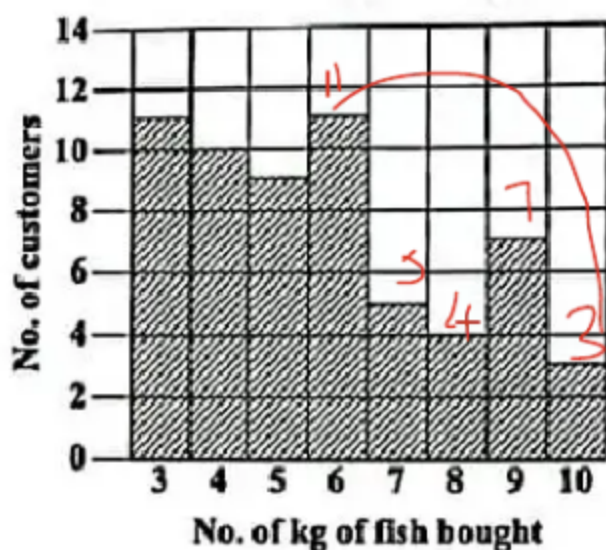
- A. 8
 - ☒ B. 9
 - C. 16
 - D. 18
- 41) The following pie chart shows the popular games played by a group of students



If 180 students play football, how many students are in the group?

- A. 300
- B. 360
- ☒ C. 720
- D. 900

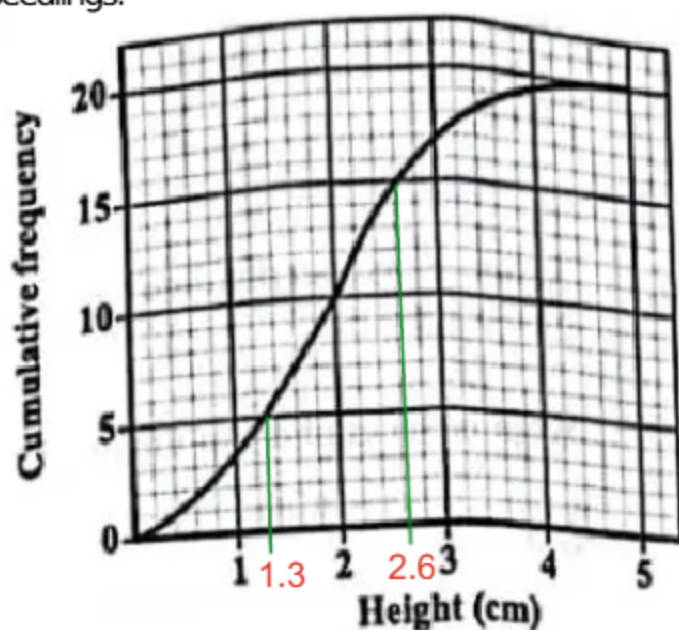
- 42) The following chart shows the amount of fish bought, in kg, by the first 60 customers at a fish market.



How many customers bought at LEAST 6kg of fish?

- A. 18
- B. 19
- ☒ C. 30
- D. 34

- 43) The following diagram shows the cumulative frequency curve of the heights of 20 seedlings.



$$\frac{1}{4} = 5^{\text{th}}$$

$$\frac{3}{4} = 15^{\text{th}}$$

$$\frac{2.6}{1.3}$$

$$\frac{1.3}{2}$$

The semi-interquartile range of the set of measurements is

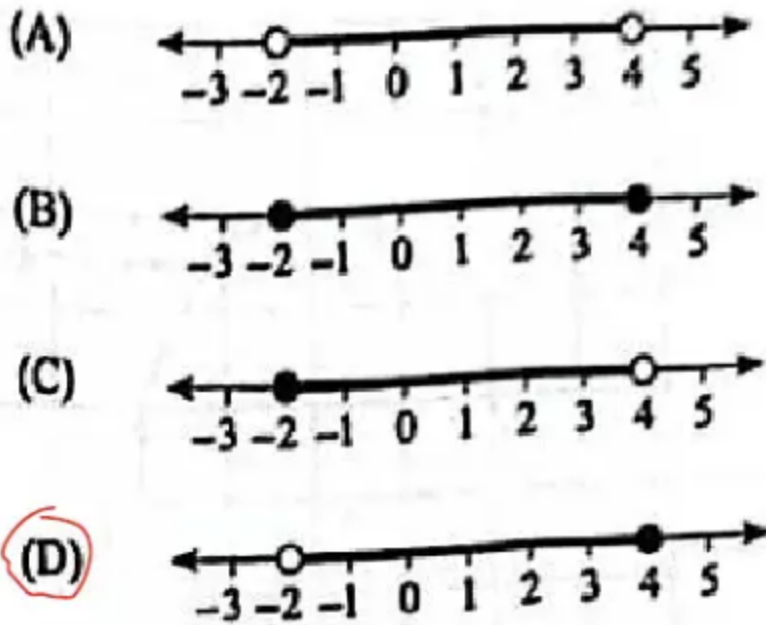
- ☒ A. 0.65 cm
- B. 1.30 cm
- C. 2.00 cm
- D. 2.60 cm

- 44) Six hundred students sit an examination. The probability of a randomly selected student failing the examination is $\frac{1}{5}$.

How many students are expected to PASS the examination?

- A. 100
- B. 120
- C. 480
- D. 500

- 45) Which of the following line graphs represents $\{x : -2 < x \leq 4\}$?



- 46) Which of the following represents the equation of a straight line?

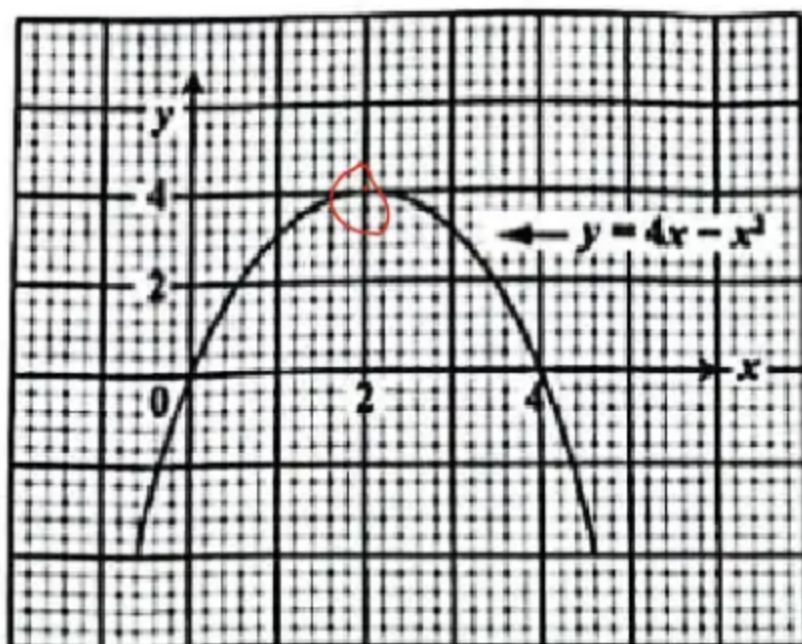
- A. $y = \frac{4}{x}$
- B. $y = 2x + 3$
- C. $y = x^2 - 4$
- D. $y = x^2 + 2x - 5$

- 47) What is the gradient of the straight line $2y = -3x - 8$?

- A. 2
- B. 3
- C. -3
- D. $-\frac{3}{2}$

$y = \underline{\hspace{2cm}}$

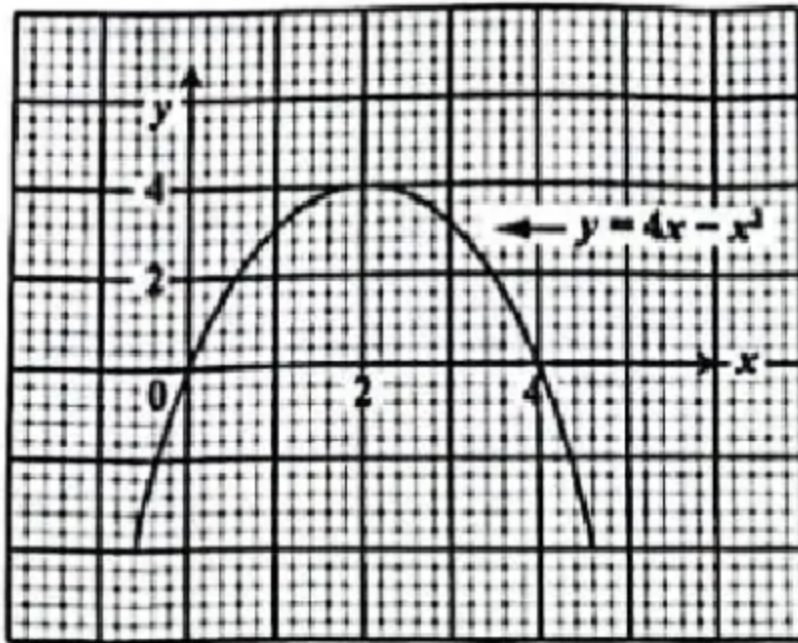
48) The following is a graph of a quadratic function.



The maximum point of $y = 4x - x^2$ is

- A. (0, 0)
- B. (0, 4)
- C. (4, 2)
- ☒ D. (2, 4)

- 49) According to the following graph, the solution of the equations $y = 4x - x^2$ and $y = 0$ are



- A. $x = 0$ when $y = 0$, and $x = 4$ when $y = 0$
 B. $x = 0$ when $y = 0$, and $x = 2$ when $y = 4$
 C. $x = 2$ when $y = 4$, and $x = 4$ when $y = 0$
 D. $x = 0$ when $y = 0$, and $x = -4$ when $y = -32$

- 50) If $h(x) = 4x^2 - 6$, then $h\left[-\frac{1}{2}\right] =$

- A. -7
 B. -5
 C. 2
 D. 5

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$$

- 51) A line L is PERPENDICULAR to the line $2x - y - 8 = 0$. What is the gradient of the line L?

- A. -2
 B. $-\frac{1}{2}$
 C. $\frac{1}{2}$
 D. 2

$$2x - 8 = y$$

$$y = 2x - 8$$

$$-1/2$$

52) Which of the following sets is represented by the function $f: x \rightarrow x^2 + 3$ where $x \in \{0, 1, 2, 3\}$?

- A. $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$
- B. $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
- C. $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
- D. $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$

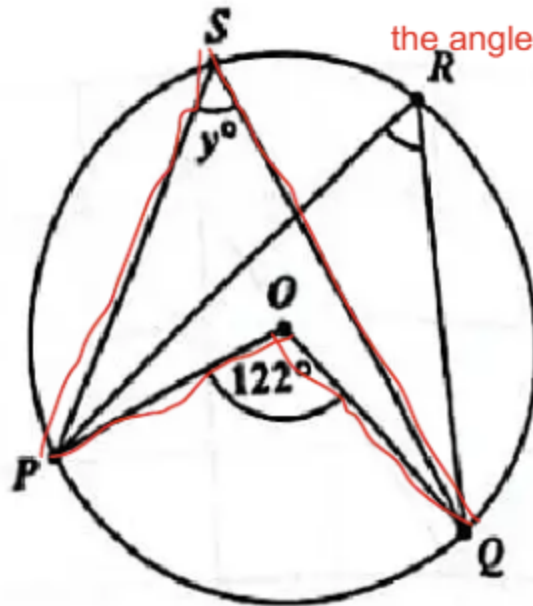
$$0^2 + 3 = 3$$

$$1^2 + 3 = 4$$

$$2^2 + 3 = 7$$

$$3^2 + 3 = 12$$

53) The following diagram is a circle with its centre at O. Angle $POQ = 122^\circ$.



the angle at the circumference is half the angle at the center

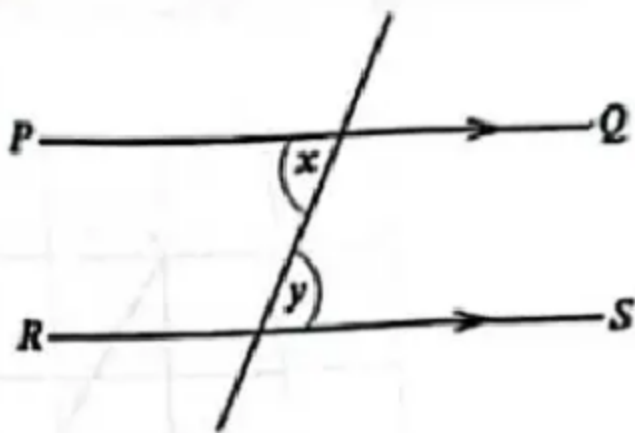
make sure the angle at the center is formed from the same 2 letters (points) as the angle at the circumference.

The value of y° is

- A. 29°
- B. 58°
- C. 61°
- D. 84°

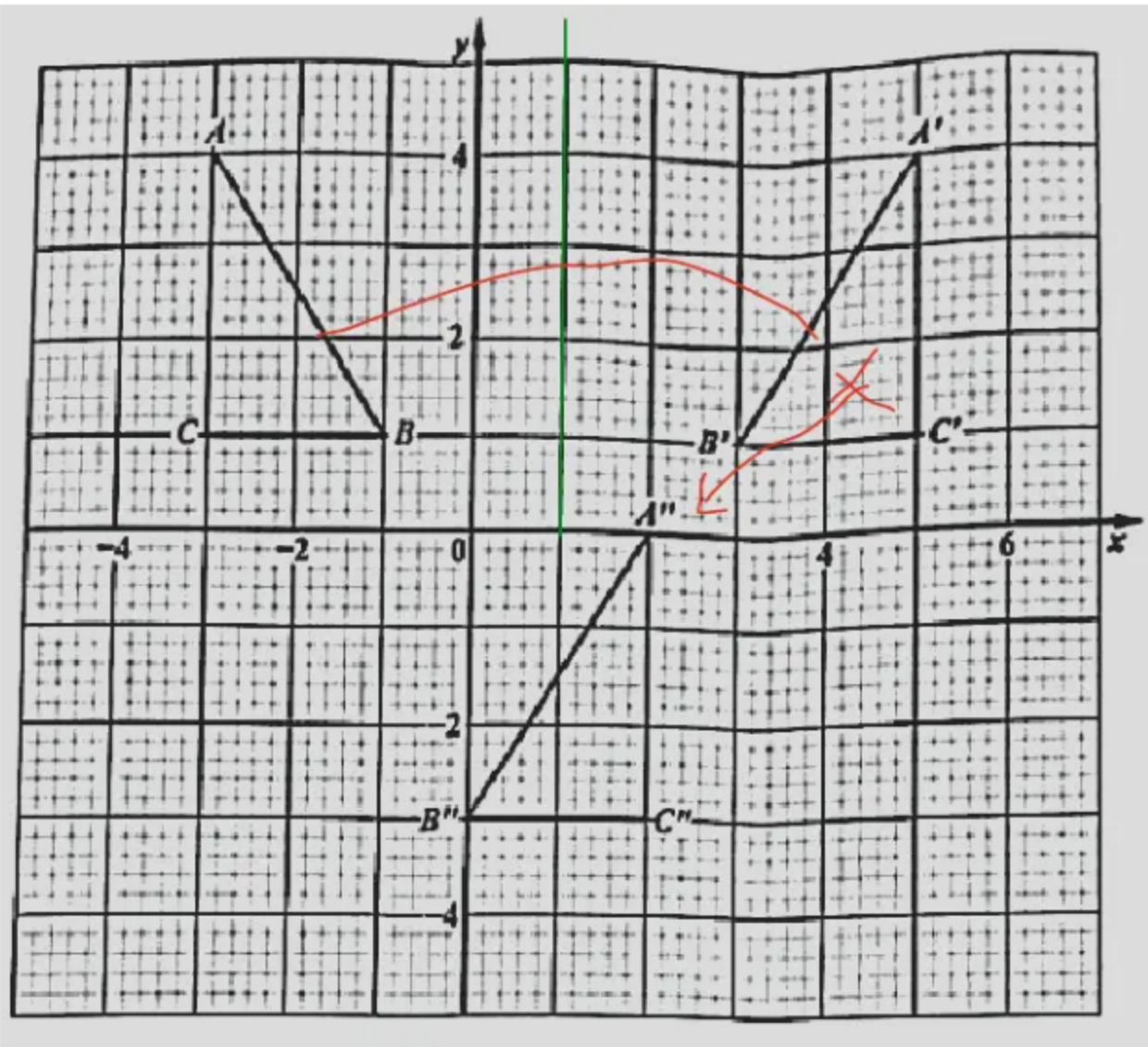
$$122/2 = 61$$

- 54 In the following diagram, PQ and RS are parallel. Which of the following BEST describes the relation between x and y ?



- A. $x = y$
B. $x < y$
C. $x + y = 90^\circ$
D. $x + y = 180^\circ$

- ✖ 55) The following diagram shows Triangle ABC and its images A'B'C' and A''B''C'' after Triangle ABC undergoes a composite/double transformation.
WWh



What sequence of transformations will map triangle ABC onto its image, triangle A''B''C''?

- ✖ A. A reflection in the line $x = 1$, followed by a translation of $\begin{bmatrix} -3 \\ -4 \end{bmatrix}$
- B. A translation of $\begin{bmatrix} -3 \\ -4 \end{bmatrix}$ followed by a reflection in the x-axis
- C. A translation of 4 units to the left, followed by a clockwise rotation of 90° about the origin
- D. A counterclockwise rotation of 90° about the origin followed by a translation of 4 units downwards

- 7) Which of the following sets is defined by $\{x \in \mathbb{Z} : -2 \leq x \leq 4\}$?
- A. $\{1, 2, 3, 4\}$
 - B. $\{0, 1, 2, 3, 4\}$
 - C. $\{-1, 0, 1, 2, 3\}$
 - ☒ D. $\{-2, -1, 0, 1, 2, 3, 4\}$

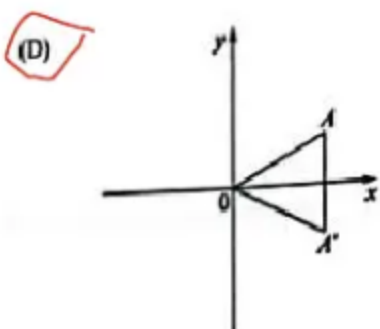
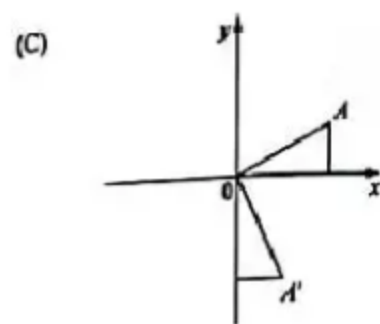
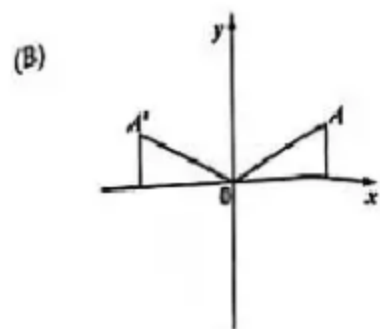
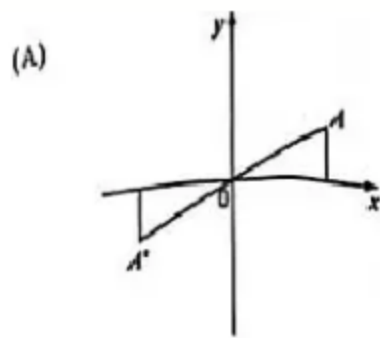
- 8) The following Venn diagram shows two intersecting sets P and Q. In the Venn diagram, $n(P) = 6$, $n(Q) = 8$ and $n(P \cap Q) = 4$. The number of elements in the shaded region of the Venn Diagram is?



$$6 + 8 = 14 - 4 = 10$$

- A. 9
 - ☒ B. 10
 - C. 14
 - D. 18
- *9) In which of the following pairs of sets is one set NOT a subset of the other?
- ☒ A. $P = \{\text{multiples of 12}\}$ and $Q = \{\text{factors of 3}\}$ $\{12, 24, 36, \dots\}$ $\{1, 3\}$
 - B. $E = \{\text{factors of 6}\}$ and $F = \{\text{factors of 12}\}$
 - C. $X = \{\text{whole numbers}\}$ and $Y = \{\text{rational numbers}\}$
 - D. $G = \{\text{multiples of five}\}$ and $H = \{\text{multiples of ten}\}$
- 10) All students in a class play Scrabble or Checkers or both. If 36% of the students play Scrabble only and 48% of the students play Checkers only, what percentage of students play BOTH games?
- A. 12
 - ☒ B. 16
 - C. 84
 - D. 88

- 56 In each of the following diagrams, A' is the image of A . which of the diagrams shows a reflection in the x -axis?

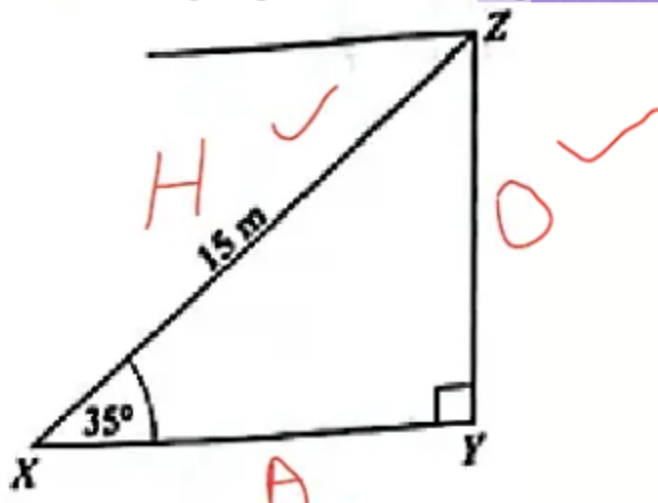


57) In triangle ABC, angle A = x° and angle B = $2x^\circ$. What is the size of angle C?

- A. $(180 - 3x)^\circ$
- B. 60°
- C. 30°
- D. $\left[\frac{180}{3x}\right]^\circ$



58) The following diagram shows the angle of elevation of a point, Z, from X.



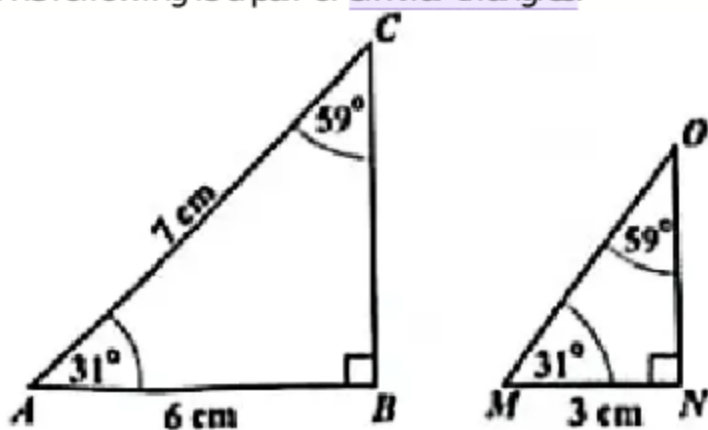
$$\sin 35 = \frac{YZ}{15}$$

$$15 \times \sin 35 = YZ$$

The angle of elevation of the point Z from X is 35° . If X is 15 metres from Z, then the height YZ, in metres, is

- A. $15 \cos 35^\circ$
- B. $15 \sin 55^\circ$
- C. $15 \sin 35^\circ$
- D. $15 \tan 55^\circ$

59) The following is a pair of similar triangles.



The length of MO, in centimetres, is

- A. 3.0
- B. 3.5
- C. 4.6
- D. 6.0

60) Under the translation $\begin{bmatrix} -2 \\ -3 \end{bmatrix}$, the image of $(-5, 3)$ is

- A. $(0, -3)$
- B. $(1, -2)$
- C. $(-7, 6)$
- D. $(3, 6)$

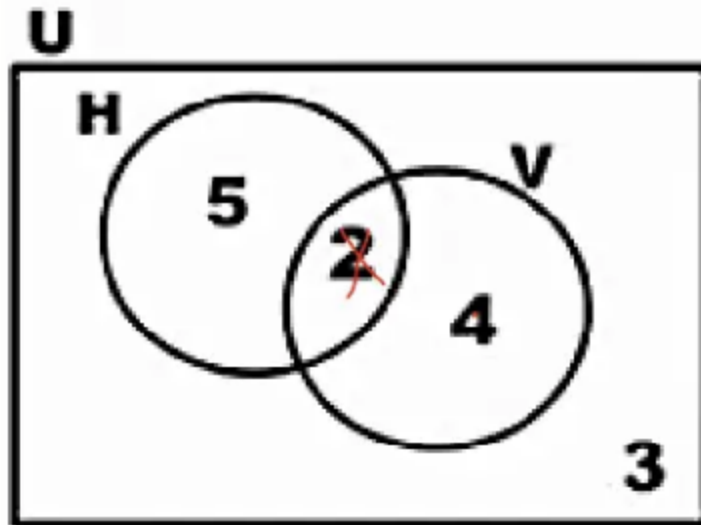
- 11) The following Venn diagram shows 2 intersecting sets. The number in each set is indicated.

In the Venn diagram,

$U = \{\text{students who play games}\}$

$H = \{\text{students who play hockey}\}$

$V = \{\text{students who play volleyball}\}$



How many students play EITHER hockey or volleyball but not both?

- A. 6
 - B. 7
 - ☒ C. 9
 - D. 11
- 12) The set of positive integers that are divisible by 7 is an example of
- A. An improper set
 - ☒ B. An infinite set
 - C. An empty set
 - D. A finite set

13) The following diagram shows part of a shopping bill.

Item	Unit Cost Price	Total Cost Price
4 kg flour	P	\$12.40
2 kg sugar	\$3.60	\$ 7.20
8 kg rice	Q	T
Subtotal		\$40.40
VAT (8%)		xxxxxx
Total		xxxxxx

20.80

The correct values of P , Q and T are

	P	Q	T
(A)	\$0.99	\$1.66	\$19.60
(B)	\$3.10	\$6.70	\$20.80
(C)	\$4.05	\$5.50	\$19.60
(D)	\$3.10	\$2.60	\$20.80

14) The following diagram shows part of a shopping bill

Item	Unit Cost Price	Total Cost Price
4 kg flour	P	\$12.40
2 kg sugar	\$3.60	\$ 7.20
8 kg rice	Q	T
Subtotal		\$40.40
VAT (8%)		xxxxxx
Total		xxxxxx

$$1\% = \div 100$$

$$40.4 \div 100 = 0.4$$

$$0.4 \times 8 = 3.2$$

According to the bill, the amount paid for VAT is

- A. \$3.10
- ☒ B. \$3.23
- C. \$37.17
- D. \$43.63

2.50 → tax

15) At a bank, EC\$2.60 is equivalent to US\$1.00. For every US \$1.00 exchanged, EC\$0.10 is deducted as an exchange tax. How many EC dollars will Leon receive if he exchanges US\$1000.00?

- A. \$900.90
- B. \$2360.34
- ☒ C. \$2500.00
- D. \$2600.00

16) A calculator which is marked at \$120 is sold for cash at a 30% discount. How much change would Susan receive if she pays for the calculator with a \$100 bill?

- ☒ A. \$16
- B. \$20
- C. \$28
- D. \$36

95%

* 17) By selling a bag for \$1140, Vishal incurred a loss of 5%. At what price should he have sold the bag to gain a profit of 5%

- A. \$1197.00
- B. \$1254.00
- C. \$1256.85
- ☒ D. \$1260.00

sell for 105%

$$1\% = 1140/95 = 12$$

$$105\% = 105 \times 12 = 1260$$

18) The cost of a machine is estimated to be decreasing at the rate of 10% every year. If it currently costs \$6000, what will be the estimated value of the machine after 2 years?

- A. \$1140
- ~~B. \$4800~~
- ☒ C. \$4860 ✓
- D. \$5400

$$\begin{aligned} \text{year 1} &= 10\% \times 6000 = 600 \\ 6000 - 600 &= 5400 \end{aligned}$$

$$\begin{aligned} \text{year 2: } 10\% \times 5400 &= 540 \\ 5400 - 540 &= 4860 \end{aligned}$$

19) A man pays 60 cents for every 200m³ of gas used, plus a fixed charge. If he pays \$178.75 when he uses 55000m³ of gas, how much is the fixed charge?

- ☒ A. \$13.75
- B. \$14.35
- C. \$151.25
- D. \$165.00

- * 20) The compound interest on \$12000 for 2 years at 10% per annum, compounded annually, is

A. \$1320
☒ B. \$2520
 C. \$13200
☒ D. \$14520

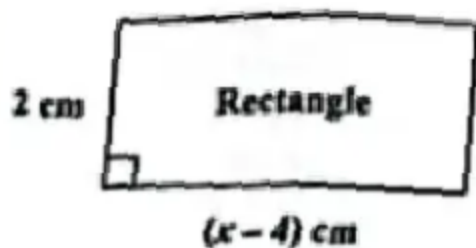
$$y1 = 10\% \times 12000 = 1200$$

$$y2 = 10\% \times 13200 = \frac{1320}{2520}$$

- 21) The product of a number p and its reciprocal maybe written as

A. $p^2 \times (-p)$
 B. $p \times (-p)$
 C. $p^2 \times \frac{1}{p}$
☒ D. $p \times \frac{1}{p}$

- 22) The following is a diagram of a rectangle



$$L \times W = x^2$$

The area of the rectangle, in cm^2 , is x^2 . the equation that maybe used to find the value of x is

☒ A. $x^2 = 2(x - 4)$
 B. $x^2 = (x - 2)(x - 4)$
 C. $x^2 = (x - 4)(x + 2)$
 D. $x^2 = 2(x - 4)(x - 2)$

- * 23) Althea, Bob and Chris collect shells. Althea has " p " shells, Bob has TWICE as many shells as Althea and Chris has 4 more than the total number of shells that both Althea and Bob collected. Altogether, the total number of shells they have is

A. $6p$
 B. $3p + 4$
 C. $6(p + 2)$
☒ D. $2(3p + 2)$

$$\begin{array}{ccc} A & B & C \\ p & 2p & 3p + 4 \end{array}$$

$$6p + 4$$

$$2(3p + 2)$$

24) If $3 + \frac{2}{x} = 1$, then the value of x is

- A. -1
- B. $\frac{1}{5}$
- C. $\frac{1}{2}$
- D. 5

* 25) The value of $5^{n+1} \times 5^{n+2}$ when $n = -1$ is

- A. 1
- B. 5
- C. 10
- D. 25

* 26) If 40 students will take 20 days to paint a wall, how many days will it take to paint the wall if 10 more students are added, working at the same rate?

- A. 12
- B. 15
- C. 16
- D. 18

1 person working for everyone = $40 \times 20 = 800$

50 person = $800 / 50 = 16$

* 27) If $P = \begin{bmatrix} 2 & 5 \\ -1 & 3 \end{bmatrix}$, then the value of $|P|$ is

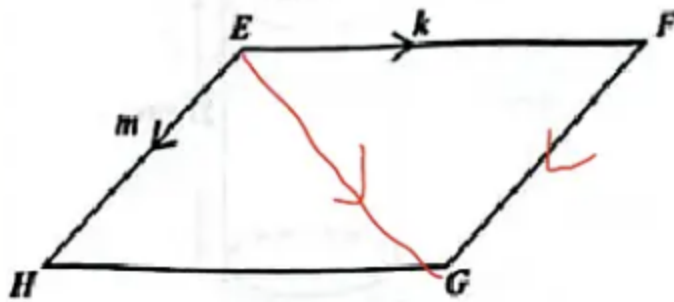
- A. 0
- B. 1
- C. 4
- D. 11

$((2 \times 3) - (5 \times -1)) = 11$

28) If $A = \begin{bmatrix} 1 & 2 & 5 & 4 \\ 6 & 1 & 3 & 7 \\ -2 & 3 & 2 & 9 \end{bmatrix}$, then the order of A is

- A. 2×3
- B. 3×2
- C. 3×4
- D. 4×3

- 29) The following is a diagram of a parallelogram, in which EF is parallel to HG, EH is parallel to FG, $\vec{EF} = k$ and $\vec{EH} = m$



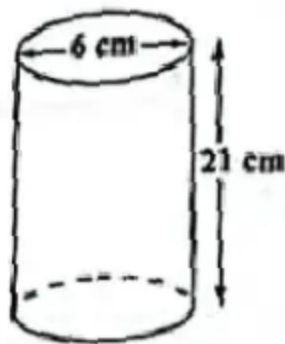
$\vec{EG}$ expressed in terms of k and m is

- A. $-m - k$
 - B. $m - k$
 - C. $k - m$
 - ☒ D. $k + m$
- 30) If $5 \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} 10 \\ 20 \end{bmatrix}$, then the values of x and y are

- A. $x = 4, y = 5$
- ☒ B. $x = 8, y = 16$
- C. $x = 2.5, y = 4$
- D. $x = 10, y = 20$

$$\begin{aligned} 5x &= 40 \\ x &= 8 \end{aligned}$$

- * 31) the following diagram shows a cylinder whose diameter is 6cm and height 21cm.



Given that $\pi = \frac{22}{7}$, then the volume of the cylinder, in cm^3 , is

- A. 120
- B. 396
- ☒ C. 594
- D. 2376

$$V = \frac{22}{7} \times 3 \times 3 \times 21$$

$$\frac{22}{7} \times \frac{9}{1} \times \frac{21}{1} = 22 \times 27$$

32) In a rectangular garden plot, 15m long and 12m wide, an area of 80m^2 is used for a vegetable garden. What area of the plot is NOT used for vegetable gardening?

- A. 26m^2
- B. 100m^2
- C. 134m^2
- D. 260m^2

* 33) How long will a speedboat take to travel between two harbours which are ^{1800km} 1808km apart, if it travels at an average speed of 120 kmh^{-1} ?

- A. 8 hours
- B. 9 hours
- C. 12 hours
- D. 15 hours

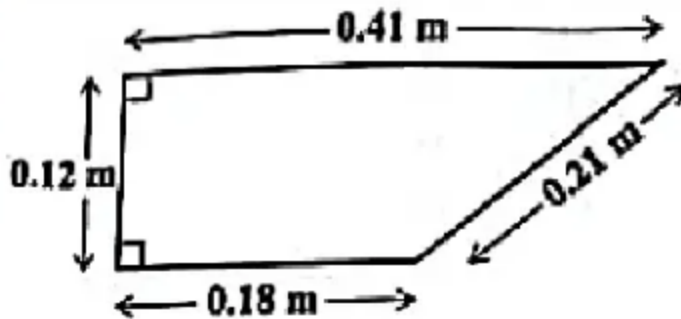
$$\begin{array}{r} 15 \\ 1800 \\ \times 120 \\ \hline \end{array}$$

34) The lengths of the sides of a triangle are x , $2x$ and $2x$ centimetres. If the perimeter is 20 centimetres, what is the value of x ?

- A. 4
- B. 5
- C. 8
- D. 10

$$\begin{aligned} 5x &= 20 \\ x &= 20/5 = 4 \end{aligned}$$

35) The following is a diagram of a trapezium.



The perimeter of the trapezium, in millimetres, is

- A. 0.092
- B. 9.2
- C. 92
- D. 920

$$\begin{array}{r} 0.41 \\ 0.12 \\ 0.18 \\ 0.21 \\ \hline \end{array}$$

$$0.92\text{m} \times 100$$